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Introduction:

The application under development is a pharmaceutical management system that eases operations in a pharmacy. Among other functions, the system comprises inventory control, sales process handling, supplier and manufacturer tracking as well as customer data retention. The main objective of this application is to enhance efficiency, accuracy, and organization within the pharmacy community.

Reason for Choosing the Application:

The decision to design a pharmaceutical management system lies in the fact that robust solutions are needed in the healthcare sector. Pharmacy stores and medical facilities are often faced with problems concerning inventory control, sales monitoring as well as legal compliances. These can be properly addressed through the use of such comprehensive software specifically designed for pharmaceutical activities. Thus, this App aims at automating manual systems reducing errors while enhancing customer service bearing in mind industry regulations, thereby increasing productivity and profitability.

Requirements Analysis:

Medicine Management:

- Users should be able to add medicine records, update them, or delete them from the system.
- Each medicine should have a scientific name, generic name, dosage, and manufacturer as designated by users.
- For every drug, the system should be capable of linking it with its manufacturer.

Supplier Management:

- Supplier records must be created added updated and deleted by the users.
- The phone numbers and names of suppliers should be recorded in the system.
- To whom each supplier has provided drugs for selling is required to be recorded by the user in this system.

Manufacturer Management:

- Information about manufacturers regarding their names and countries of origin should be stored at the system/organization level for regulatory purposes.
- Manufacturer records may have to be added, updated, or even deleted from time to time by the users in the organization.
- Users should be able to link each manufacturer to the medicines they produce.

Inventory Management:
- Add, update, and delete inventory records supported by the system.
- For all inventory items, purchase price, expiry date, and quantity need to be inputted into the application area where required.
- Automatic notifications will go out when there are low stock levels or items that are near the expiration date.

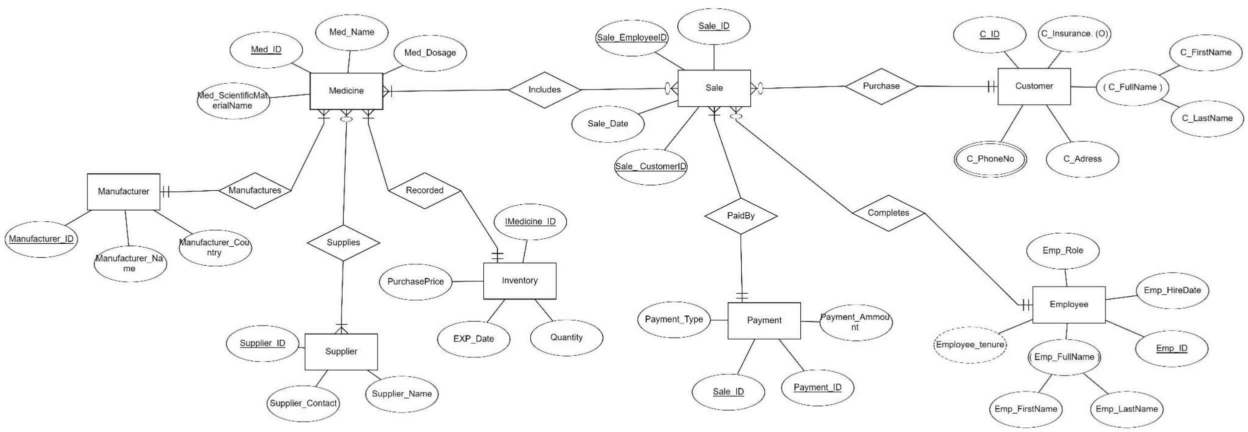
Sale Management:
- Users/clients/patients need access to sales processing options involving picking drugs from the stock list, placing them in sales baskets/carts or bags, and calculating total costs/amounts.
- Customer information, medicines sold, payment method, and date/time of sale will be captured by the system during each transaction.

Customer Management:
- Customers should be capable of managing clients' records including private details, insurance information, and contacts.
- New customers can be added, existing records can be updated, and the system makes searching for customers easy; while searching for clients with different standards.
- For personalized services and marketing purposes, customer purchase history must be kept.

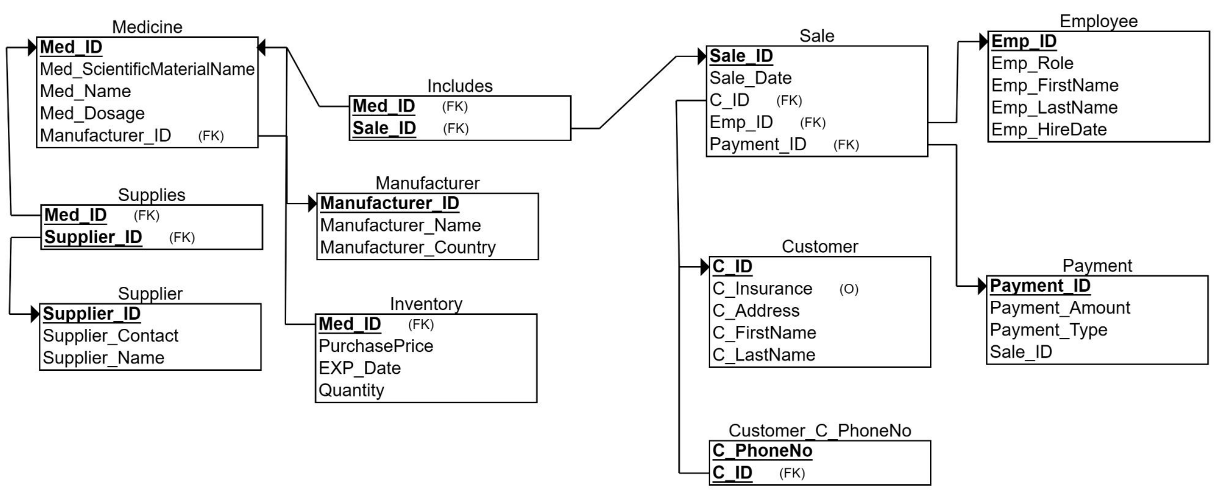
Customer Phone Number Management:
- Customer phone numbers should be managed by users and connected with client profiles.
- This system should allow adding, updating, and deleting customer phone numbers.

Employee Management:
- Employees' records such as role, name, hire date, and contact details should be properly managed by users of the system.
- Access to sensitive functions should only be done through role-based access control, depending on their roles in an organization which restricts them from accessing those functions that are sensitive.

This ER+EER diagram effectively captures the entities, attributes, and relationships required for the pharmaceutical management system:



This relational schema represents the tables and their respective attributes derived from the ER diagram:



This sample database instance provides an illustration of how the data is structured within each table:

Supplier:

SUPPLIER_ID	SUPPLIER_CONTACT	SUPPLIER_NAME
1111	791234567	Ahmad
1112	791563565	Mohammad
1113	791974658	Yousef
1114	786246836	Dana
1115	791293745	Lara

Manufacturer:

MANUFACTURER_ID	MANUFACTURER_NA	MANUFACTURER_CO
111	Hikma	Jordan
112	JIP	Jordan
121	HJJ	Jordan
113	PIW	Jordan
131	BNP	Jordan

Medicine:

MED_ID	MED_SCIENTIFICMATERIALNAME	MED_NAME	MED_DOSAGE	MANAFACTURER_ID
23	Hydrochloric Acid	Cortrope	70	112
35	Acetic Anhydride	Amoclan	200	121
18	Benzyl Chloride	Panadol	500	131
12	Diethyl Chloride	Sinecode	100	111
29	Acetone	Allegra	2000	112

Supplies:

MED_ID	SUPPLIER_ID
23	1112
35	1111
18	1113
12	1114
29	1115

Inventory:

MED_ID	PURCHASEPRICE	EXP_DATE	QUANTITY
23	1.5	01-JAN-25	209
35	5	01-FEB-26	26
18	2.2	01-MAY-25	798
12	2	01-DEC-25	1500
29	7.5	01-MAR-27	120

Customer:

C_ID	C_I	C_ADDRESS	C_FIRSTNAME	C_LASTNAME
11111	YES	AMMAN	Ahmad	Karadsheh
11112	NO	MADABA	Omar	Hammad
11113	YES	AMMAN	Khalid	Alshareef
11123	YES	AQABA	Maryam	Abudagga
11133	NO	IRBID	Lujain	Kofahi

Employee:

EMP_ID	EMP_ROLE	EMP_FIRSTNAME	EMP_LASTNAME	EMP_HIRED
22222	Pharamcist	Amr	Hasan	20-JAN-24
22221	Manager	Waseem	Ramez	17-NOV-23
22223	Pharmacist	Abdullah	Melhem	04-MAY-23
22224	Accountant	Laith	Masaeed	06-JUN-22
22225	Pharmacist	Osama	Qubbaj	04-JUL-22

Payment:

PAYMENT_ID	PAYMENT_AMOUNT	PAYM	SALE_ID
33333	2	VISA	90002
33332	3	VISA	90005
33339	5	CASH	90003
33343	9	CASH	90001
33331	8	VISA	90007

CREATE:

```
SQL> CREATE TABLE Supplier(
Supplier_ID numeric(4) NOT NULL,
Supplier_Contact numeric(10) NOT NULL,
Supplier_Name VARCHAR(15),
PRIMARY KEY (Supplier_ID)
);

Table created.

SQL>
SQL> CREATE TABLE Manufacturer(
Manufacturer_ID numeric(3),
Manufacturer_Name VARCHAR(15) NOT NULL,
Manufacturer_Country VARCHAR(15),
PRIMARY KEY (Manufacturer_ID)
);

Table created.

SQL>
SQL> CREATE TABLE Medicine(
Med_ID numeric(2),
Med_ScientificMaterialName VARCHAR(30) NOT NULL,
Med_Name VARCHAR(30) NOT NULL,
Med_Dosage numeric(5),
Manafacturer_ID numeric(3) NOT NULL,
PRIMARY KEY (Med_ID),
FOREIGN KEY (Manafacturer_ID) REFERENCES Manufacturer
(Manufacturer_ID)
);

Table created.
```

```
SQL> CREATE TABLE Supplies(
  2  Med_ID numeric(2) NOT NULL,
  3  Supplier_ID numeric(4) NOT NULL,
  4  FOREIGN KEY (Med_ID) REFERENCES Medicine(Med_ID),
  5  FOREIGN KEY (Supplier_ID) REFERENCES Supplier(Supplie
r_ID)
  6  );

Table created.

SQL>
SQL> CREATE TABLE Inventory(
  2  Med_ID numeric(2),
  3  PurchasePrice numeric(3,2) NOT NULL,
  4  EXP_Date Date NOT NULL,
  5  Quantity numeric(4),
  6  FOREIGN KEY (Med_ID) REFERENCES Medicine(Med_ID)
  7  );

Table created.

SQL>
SQL> CREATE TABLE Customer(
  2  C_ID numeric(5),
  3  C_Insurance VARCHAR(3),
  4  C_Address VARCHAR(10),
  5  C_FirstName VARCHAR(15),
  6  C_LastName VARCHAR(15),
  7  PRIMARY KEY (C_ID)
  8  );

Table created.
```

```
SQL>
SQL> CREATE TABLE Employee(
  2  Emp_ID numeric(5),
  3  Emp_Role VARCHAR(10),
  4  Emp_FirstName VARCHAR(15),
  5  Emp_LastName VARCHAR(15),
  6  Emp_HireDate Date,
  7  PRIMARY KEY (Emp_ID)
  8  );

Table created.

SQL>
SQL> CREATE TABLE Payment(
  2  Payment_ID numeric(5),
  3  Payment_Amount numeric(3,2),
  4  Payment_Type VARCHAR(4),
  5  Sale_ID numeric(6),
  6  PRIMARY KEY (Payment_ID)
  7  );

Table created.

SQL>
SQL> CREATE TABLE Sale(
  2  Sale_ID numeric(6),
  3  Sale_Date Date,
  4  C_ID numeric(5),
  5  Emp_ID numeric(5),
  6  Payment_ID numeric(5),
  7  PRIMARY KEY (Sale_ID),
  8  FOREIGN KEY (C_ID) REFERENCES Customer(C_ID),
  9  FOREIGN KEY (Emp_ID) REFERENCES Employee(Emp_ID),
 10  FOREIGN KEY (Payment_ID) REFERENCES Payment(Payment_I
D)
 11  );

Table created.
```

```
SQL> CREATE TABLE Includes(
     Med_ID numeric(2),
     Sale_ID numeric(6),
     FOREIGN KEY (Med_ID) REFERENCES Medicine(Med_ID),
     FOREIGN KEY (Sale_ID) REFERENCES Sale(Sale_ID)
     );

Table created.

SQL>
SQL> CREATE TABLE Customer_C_PhoneNo(
     C_PhoneNo numeric(10),
     C_ID numeric(5),
     PRIMARY KEY (C_PhoneNo),
     FOREIGN KEY (C_ID) REFERENCES Customer(C_ID)
     );

Table created.
```

INSERT:

```
SQL> INSERT INTO Supplier VALUES(1111,0791234567,'Ahmad');


1 row created.

SQL> INSERT INTO Supplier VALUES(1112,0791563565,'Mohammad
');

1 row created.

SQL> INSERT INTO Supplier VALUES(1113,0791974658,'Yousef')
;

1 row created.

SQL> INSERT INTO Supplier VALUES(1114,0786246836,'Dana');

1 row created.

SQL> INSERT INTO Supplier VALUES(1115,0791293745,'Lara');

1 row created.

SQL> INSERT INTO Manufacturer VALUES(111,'Hikma','Jordan')
;

1 row created.

SQL> INSERT INTO Manufacturer VALUES(112,'JIP','Jordan');

1 row created.

SQL> INSERT INTO Manufacturer VALUES(121,'HJJ','Jordan');

1 row created.

SQL> INSERT INTO Manufacturer VALUES(113,'PIW','Jordan');

1 row created.
```

```
SQL> INSERT INTO Manufacturer VALUES(131,'BNP','Jordan');

1 row created.

SQL> INSERT INTO Medicine VALUES (23,'Hydrochloric Acid','
Cortrope',70,112);

1 row created.

SQL> INSERT INTO Medicine VALUES (35,'Acetic Anhydride','A
moclan',200,121);

1 row created.

SQL> INSERT INTO Medicine VALUES (18,'Benzyl Chloride','Pa
nadol',500,131);

1 row created.

SQL> INSERT INTO Medicine VALUES (12,'Diethyl Chloride','S
inecode',100,111);

1 row created.

SQL> INSERT INTO Medicine VALUES (29,'Acetone','Allegra',2
000,112);

1 row created.

SQL> INSERT INTO Supplies VALUES (23,1112);

1 row created.

SQL> INSERT INTO Supplies VALUES (35,1111);

1 row created.

SQL> INSERT INTO Supplies VALUES (18,1113);

1 row created.
```

```
SQL> INSERT INTO Supplies VALUES (12,1114);

1 row created.

SQL> INSERT INTO Supplies VALUES (29,1115);

1 row created.

SQL> INSERT INTO Inventory VALUES (23,1.5,'01/jan/2025',20
9);

1 row created.

SQL> INSERT INTO Inventory VALUES (35,5,'01/feb/2026',26)
  2  ;

1 row created.

SQL> INSERT INTO Inventory VALUES (18,2.2,'01/may/2025',79
8);

1 row created.

SQL> INSERT INTO Inventory VALUES (12,2,'01/dec/2025',1500
);

1 row created.

SQL> INSERT INTO Inventory VALUES (29,7.5,'01/mar/2027',12
0);

1 row created.

SQL> INSERT INTO Customer VALUES (11111,'YES','AMMAN','Ahm
ad','Karadsheh');

1 row created.

SQL> INSERT INTO Customer VALUES (11112,'NO','MADABA','Oma
r','Hammad');
```

```
SQL> INSERT INTO Customer VALUES (11113,'YES','AMMAN','Kha
lid','Alshareef');

1 row created.

SQL> INSERT INTO Customer VALUES (11123,'YES','AQABA','Mar
yam','Abudagga');

1 row created.

SQL> INSERT INTO Customer VALUES (11133,'NO','IRBID','Luja
in','Kofahi');

1 row created.
```

```
SQL> INSERT INTO Employee VALUES (22222,'Pharamcist','Amr'
,'Hasan','20/jan/2024');

1 row created.

SQL> INSERT INTO Employee VALUES (22221,'Manager','Waseem'
,'Ramez','17/nov/2023');

1 row created.

SQL> INSERT INTO Employee VALUES (22223,'Pharmacist','Abdu
llah','Melhem','04/may/2023');

1 row created.

SQL> INSERT INTO Employee VALUES (22224,'Accountant','Lait
h','Masaeed','06/jun/2022');

1 row created.

SQL> INSERT INTO Employee VALUES (22225,'Pharmacist','Osam
a','Qubbaj','04/jul/2022');

1 row created.
```

```
SQL> INSERT INTO Payment VALUES (33332,3,'VISA',90005);

1 row created.

SQL> INSERT INTO Payment VALUES (33339,5,'CASH',90003);

1 row created.

SQL> INSERT INTO Payment VALUES (33343,9,'CASH',90001);

1 row created.

SQL> INSERT INTO Payment VALUES (33331,8,'VISA',90007);

1 row created.

SQL> INSERT INTO Sale VALUES (90002,'04/may/2024',11111,22
223,33333);

1 row created.

SQL> INSERT INTO Sale VALUES (90005,'01/may/2024',11113,22
224,33332);

1 row created.

SQL> INSERT INTO Sale VALUES (90001,'29/apr/2024',11123,22
221,33339);

1 row created.

SQL> INSERT INTO Sale VALUES (90003,'12/mar/2024',11112,22
222,33343);

1 row created.

SQL> INSERT INTO Sale VALUES (90007,'14/nov/2023',11133,22
225,33331);
```

```
SQL> INSERT INTO Includes VALUES (23,90001);

1 row created.

SQL> INSERT INTO Includes VALUES (35,90003);

1 row created.

SQL> INSERT INTO Includes VALUES (18,90002);

1 row created.

SQL> INSERT INTO Includes VALUES (29,90007);

1 row created.

SQL> INSERT INTO Includes VALUES (12,90005);

1 row created.

SQL> INSERT INTO Customer_C_PhoneNo VALUES (0792837235,111
11);

1 row created.

SQL> INSERT INTO Customer_C_PhoneNo VALUES (0792837215,111
12);

1 row created.

SQL> INSERT INTO Customer_C_PhoneNo VALUES (0792837225,111
23);

1 row created.

SQL> INSERT INTO Customer_C_PhoneNo VALUES (0792837284,111
13);

1 row created.
```

```
SQL> Update Payment set Payment_Amount=2 WHERE Payment_ID=33333;

1 row updated.

SQL> Delete from Inventory where Med_ID=23;

1 row deleted.

SQL> SELECT S.Sale_ID,P.Payment_ID from Sale S JOIN Payment P ON S.Payment_ID=P.Payment_ID;

   SALE_ID PAYMENT_ID
---------- ----------
     90002      33333
     90005      33332
     90001      33339
     90003      33343
     90007      33331

SQL> Select count(C.C_ID),C.C_Insurance from Customer C JOIN Sale S ON C.C_ID=S.C_ID
  2   where C.C_FirstName='Omar' GROUP BY C.C_Insurance;

COUNT(C.C_ID) C_I
------------- ---
            1 NO
```